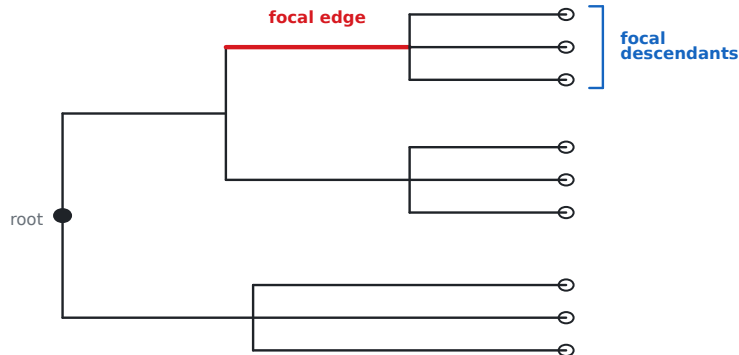


a One branch, one site



one site

G
G
G

Focal clade: G G G

Outside includes G

A
A
A

A
A
G

b Two questions, different answers

OBSERVED

Clade-exclusive?

focal	G	G	G			
outside	A	A	A	A	A	G

NO

not fixed-exclusive

G occurs outside the clade

≠

kept separate

INFERRED

Substitution on the focal edge?



YES

unambiguous A→G

Only optimal focal-edge pair: A→G

c Retain all optimal focal-edge state pairs

Optimal parent→child pair(s)

A→G only

A→G or C→G

A→G or G→G

G→G only

BRANCHSNV reports

Unambiguous change

State ambiguity

Placement ambiguity

No change

No arbitrary tie-breaking

All globally optimal focal-edge state pairs are retained.

DESIGN SAFEGUARDS

Explicit root
defines direction

Exact descendant set
defines branch

Exact taxon labels
define correspondence

Complete optimal set
retains uncertainty

Deterministic provenance
supports auditability